



Blitz Mock Series — #9

Topper Notes — All 25 Questions | Complete Explanation

Physics · Geography · Chemistry · Polity · Soil · Biology · Sun · Cell



→ Join Telegram: t.me/Exam_Zen

① Physics — Sound Wave: Air → Water

Q1. A sound wave travels from air into water. How do its speed and wavelength change?

- ☐ A ~~Speed increases, Wavelength decreases~~
- ☐ B ~~Speed decreases, Wavelength increases~~
- ☒ C **Speed increases, Wavelength increases ✓**
- ☐ D ~~Speed decreases, Wavelength decreases~~

✓ Answer: C — Air → Water = denser medium → Sound speeds UP → Wavelength also increases ($v = f\lambda$, f is constant)

→ Nature of Sound Waves

- ✓ **Longitudinal waves** → particles vibrate parallel to direction of propagation (same as seismic P-waves)
- ✓ **Mechanical waves** → require a medium | Cannot travel in vacuum
- ✓ **Cannot be polarized** (only transverse waves can be polarized)

FORMULA

Speed of Sound in Media: **Solid > Liquid > Gas**

Key formula: $v = f \times \lambda$ | Frequency (f) depends only on **source** — stays constant during medium change

$\therefore$ Air $\rightarrow$ Water: $v \uparrow + f$ unchanged $\rightarrow \lambda \uparrow$ (wavelength increases)

→ Sound vs Light — Denser Medium Comparison

Property	Sound (Longitudinal)	Light (Transverse / EM Wave)
Air $\rightarrow$ Water (Speed)	INCREASES $\uparrow$	DECREASES $\downarrow$
Wavelength change	Increases	Decreases
Frequency	Unchanged	Unchanged
Needs medium?	YES	No (EM wave)
Polarizable?	No	Yes

EXAM TRAP!

Sound in denser medium = FASTER | **Light in denser medium = SLOWER** — ye ulta hai! Confuse mat hona!

→ Refraction of Light (Snell's Law)

LAWS OF REFRACTION

1. Incident ray, refracted ray, normal $\rightarrow$ all in **same plane**

2. $\sin i / \sin r = \mu$ (refractive index = constant)

Rarer $\rightarrow$ Denser: bends **toward normal**

Denser $\rightarrow$ Rarer: bends **away from normal**

LAWS OF REFLECTION

1. Incident, reflected, normal $\rightarrow$ **same plane**

2. $\angle i = \angle r$ (angle of incidence = reflection)

Echo = reflection of sound

Min. dist for echo: **17.2 metres**

Repeated reflection = **Reverberation**

MEMORY TRICK

Maihar Gharana: Alauddin Khan (Founder) $\rightarrow$ Ali Akbar Khan (Sarod, son) + Annapurna Devi (Surbahar, daughter, 1st wife of Ravi Shankar) + Pandit Ravi Shankar (Sitar, son-in-law)

② Geography — Atmospheric Layers

Q2. Which of the following is a characteristic of the Thermosphere?

- ☐ A ~~It is the coldest layer of the atmosphere~~
- ☐ B ~~It contains the most atmospheric mass~~
- ☒ C **Its temperature can rise above 2000°C ✓**
- ☐ D ~~Meteors burn up in this layer~~

✓ Answer: C — Thermosphere temp can exceed 2000°C | Coldest = Mesosphere | Meteors burn in Mesosphere

→ Five Atmospheric Layers — Complete Table

Layer	Height	Temp Pattern	Key Features
Troposphere	0–13 km avg (8 km poles; 18 km equator)	Decreases ↓	All weather; most gases; lapse rate 6.5°C/km
Stratosphere	13–50 km	Increases ↑	Ozone layer; jet planes fly here; no weather
Mesosphere	50–85 km	Decreases ↓	Coldest layer; Meteors burn here
Thermosphere	85 km+	Increases ↑ (2000°C+)	Ionosphere; reflects radio waves; ions present
Exosphere	Outermost	—	Outermost layer

MEMORY TRICK

Temperature Pattern: ↓ Tropo → ↑ Strato → ↓ Meso → ↑ Thermo

M = M trick: Mesosphere = Meteors burn here!

→ Greenhouse Effect — Why Troposphere cools going up

- GHGs absorb **outgoing terrestrial radiation** (long wavelength) — transparent to incoming solar radiation (short wavelength)
- Surface gets heated → radiates long wave → absorbed near ground → heat stays near surface

KEY FACTS

Greenhouse gases (most to least prominent): **Water Vapour** → **CO₂** → **Methane** →

Nitrous oxide

Troposphere extends higher at equator (18 km) vs poles (8 km) → stronger convection currents at equator

Thermosphere = Ionosphere → **reflects radio waves** → enables long-distance radio communication

♦ ♦ ♦

③ Chemistry — Strong Acid in Water

Q3. Which of the following is a strong acid, completely ionizing to release H⁺ ions?

☒ A HCl (Hydrochloric acid) ✓

☐ B CH₃COOH (Acetic acid)

☐ C H₂CO₃ (Carbonic acid)

☐ D H₃PO₄ (Phosphoric acid)

✓ Answer: A — HCl is a **STRONG** acid (complete ionization) | Others are weak acids (partial ionization)

EXAM TRAP!

Acid Dilution Rule: Always add **ACID to WATER** — never water to acid!

Reason: Reaction is **highly exothermic** → adding water to acid causes violent splashing, burns, blisters

→ Strong vs Weak Acids

Acid	Type	Ionization	Source
HCl	Strong	Complete	—
CH ₃ COOH (Acetic acid)	Weak	Partial	Vinegar (5–8%)
H ₂ CO ₃ (Carbonic acid)	Weak	Partial	Carbonated drinks
H ₃ PO ₄ (Phosphoric acid)	Weak	Partial	—

→ Bronsted-Lowry Concept + Neutralization

BRONSTED-LOWRY

Acid = H^+ donor

Base = H^+ acceptor

$HCl \rightarrow$ donates $H^+ \rightarrow$ becomes Cl^-

(conjugate base)

$H_2O \rightarrow$ accepts $H^+ \rightarrow$ becomes H_3O^+

(conjugate acid)

NEUTRALIZATION

Acid + Base $\rightarrow$ Salt + Water

Strong + Strong $\rightarrow$ Neutral (pH 7)

Strong acid + Weak base $\rightarrow$ Acidic (pH < 7)

Weak acid + Strong base $\rightarrow$ Basic (pH > 7)



④ Modern History — Kuka Movement 🏆

Q4. Who founded the Kuka Movement in Punjab?

☐ A Jawaharmal

☒ B Namdhari Guru Ram Singh ✓

☐ C Lala Lajpat Rai

☐ D Bhagat Singh

✓ Answer: B — Guru Ram Singh Ji (Baba Ram Singh) | Also called Namdhari Movement | Founded 1857

Detail	Information
Movement Name	Kuka Movement
Also known as	Namdhari Movement
Founded	1857
Founded by	Guru Ram Singh Ji (Namdhari / Baba Ram Singh)
Purpose	To purify Sikhism from corruption
Note	Jawaharmal ignited it first but Guru Ram Singh propounded and led it



⑤ Polity — Constitutional Borrowings: CAG ⚖️

Q5. In our Constitution, the office of CAG was borrowed from which country's constitution?

☒ A Britain ✓

☐ B Australia

☐ C Canada

☐ D USA

✓ Answer: A — CAG borrowed from Britain (Westminster model) | Part V, Chapter 5, Articles 148–151

→ Constitutional Borrowings — Country-wise

Country	What was borrowed
Britain (UK)	CAG, Parliamentary form of Govt (Westminster), Single citizenship, Bicameral legislature, Rule of Law, Speaker in Lok Sabha, Martial Law
Australia (SLT)	S= Joint Sitting, L= Concurrent List, T= Trade & Commerce provisions, Doctrine of Repugnancy (Art.254)
Canada (3 Cs)	Federation with strong Centre , Centre appoints Governor , Residuary powers with Centre
USA	Fundamental Rights, Judicial Review, Judicial Independence, President & VP, Impeachment, Removal of SC/HC judges
South Africa	Amendment procedure (Article 368, Part XX)
Ireland	RS nominated members, DPSP, Election of President

→ CAG — Complete Details

CAG FACTS

Constitutional provision: **Part V, Chapter 5**

Articles: **148 to 151**

Art.148 → Appointment (by President)

Art.149 → Duties & powers

Art.150 → Forms of accounts

Art.151 → Reports to President

KEY DETAILS

First CAG: **V. Narahari Rao**

Current CAG: **Sanjay Murthy**

Title: **Guardian of the Public Purse**

Role: Guide & philosopher of **PAC**

→ Parliamentary Committees

Committee	Members	Formed
Public Accounts Committee (PAC)	22 (15 LS + 7 RS)	Gol Act 1919; formed 1921

KEY FACTS

- Joint Sitting (Art.108): Rules of Lok Sabha apply | Presided by **Speaker of Lok Sabha** | Passed by **simple majority**
- Joint Sitting only for **Ordinary Bills** — NOT Money Bill, NOT Constitution Amendment Bill
- Joint Sitting happened only **3 times** so far



⑥ Art & Culture — Temple Architecture

Q6. Stmt 1: Amalaka sits atop Shikhara in Nagara style. | Stmt 2: In Dravidian style, Vimana refers to gateway tower (Gopuram).

- ☐ A Both correct
- ☒ B Statement 1 correct, Statement 2 incorrect ✓
- ☐ C Statement 1 incorrect, Statement 2 correct
- ☐ D Both incorrect

✓ Answer: B — Amalaka atop Shikhara ✓ | Vimana ≠ Gopuram (they are TWO different structures) ✗

→ Three Main Temple Architecture Styles

Style	Region	Developer
Nagara	North India	—
Dravidian	South India	—
Vesara	Central India (Maharashtra, Karnataka)	Chalukyas

→ Nagara Style — Components (Bottom to Top)

NAGARA BOTTOM → TOP

Jagati (raised platform) → **Garbhagriha** (sanctum sanctorum) → **Mandapa** (pillared hall)
 → **Shikhara** (curvilinear tower) → **Amalaka** (ribbed disc/gooseberry atop Shikhara) → **Kalasha** (pot finial atop Amalaka)

EXAM TRAP!

Dravidian Style: **Vimana** = pyramidal tower ABOVE sanctum (NOT gateway!)

Gopuram = gateway towers at entrance — completely DIFFERENT from Vimana!

→ Regional Variations

Region	Style	Examples / Key Features
Odisha	Kalinga	Konark Sun Temple, Jagannath Puri, Lingaraja Temple
Karnataka	Hoysala	Star-shaped/Stellar plan Hoysaleswara (Shiva) + Chennakeshava (Vishnu)
Western India	Maru-Gurjara	—



⑦ Sports — Taekwondo Matching

Q7. Match Taekwondo terms: Tae / Kwon / Do / Dojang

☐ A ~~A=B, B=C, C=A, D=D~~

☒ B **C-B-A-D ✓**

☐ C ~~B-C-D-A~~

☐ D ~~A-B-C-D~~

✓ Answer: B — Tae=Feet(C) | Kwon=Hand(B) | Do=Discipline/Way(A) | Dojang=Training Hall(D)

Term	Meaning
Tae	Strike with Feet
Kwon	Strike with Hand
Do	Discipline / Way
Dojang	Training Hall / Place of Practice

💡 EXAM TIP

Matching Question Strategy: First identify **common option-pairs** across choices → 2 eliminated immediately → use eliminated pairs for hints → works in 95% matching questions even without knowledge!

⑧ Geography — Soils of India

Q8. Which soil is deficient in nutrients and mainly found in tropical/subtropical climates?

- ☐ A Red Soil
- ☐ B Arid Soil
- ☐ C Black Soil (Regur)
- ☒ D Laterite Soil ✓

✓ Answer: D — Laterite Soil | Heavy rainfall → Leaching (nichalan) → nutrients washed away → nutrient-poor

→ All Major Soil Types — Comparison

Soil	Location	Key Features	Best Crop
Red Soil	Leeward side Western Ghats; water-deficient areas	Iron oxide → red color; less rainfall areas	General crops
Arid Soil	Desert areas (very low rainfall)	Saline, infertile	—
Black Soil (Regur)	NW Deccan Plateau (Maharashtra, Gujarat, SW MP)	Volcanic origin; self-plowing; cracks when dry	Cotton (210 frost-free days)
Laterite Soil	Tropical/subtropical heavy rainfall areas	Leaching removes silica & nutrients → iron-rich	Cashew, Coffee, Tea
Alluvial Soil	Northern Plains	Most fertile; Bhangar (old) + Khadar (new)	Multiple crops

LEACHING PROCESS

Heavy rainfall → water percolates

Leaching (Nichalan) = minerals

including silica washed downward

Result → nutrient-deficient, iron-rich

laterite soil

SOIL HORIZONS (O-A-B-C-R)

O = Organic/Humus (dark; excellent fertility)

A = Topsoil (most fertile)

B = Illuviation zone

C = Weathered rock | **R** = Bedrock

Study of soil = **Pedology**

KEY FACTS

- Soil Macronutrients: **NPK + CHO + CaMgS** = 9 total
- Khadar is more fertile than Bhangar (renewed regularly by floods)



⑨ Polity — Fundamental Rights: Article 33

Q? What is Article 33 of Part Three of the Constitution about?

- ☐ A Right to property for all citizens
- ☒ B Parliament can restrict/modify Fundamental Rights for Armed Forces ✓
- ☐ C Special provisions for women and children
- ☐ D Abolition of untouchability

✓ Answer: B — Art.33: Parliament (NOT State Legislature) can restrict FR for Armed Forces

→ Part III — Important Articles (12 to 35)

Article	Content
12	Definition of State (Parliament, Gol, State Govts, PSUs, LIC — BCCI excluded)
13	Laws inconsistent with FR = void ; source of Judicial Review
17	Abolition of Untouchability
19	Six freedoms (only for citizens)
32	Right to Constitutional Remedies
33	Parliament can restrict/modify FR for Armed Forces (NOT State Legislature)
34	Martial law provisions
35	Only Parliament (not State) can make laws regarding FRs

→ Article 15 — Prohibition of Discrimination (Clauses)

Clause	Content	Added By
Clause 3	State CAN make special provisions for women and children	Original
Clause 4	Reservation for backward classes/SC/ST permitted	1st Constitutional Amendment (post Champakam Dorairajan Case 1951)

Clause 6	EWS reservation permitted	Recent amendment
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ONLY FOR CITIZENS (not foreigners)

Articles: 15, 16, 19, 29, 30

CANNOT BE SUSPENDED (Emergency)

Articles: 20 and 21

KEY FACTS

🔑 Six Fundamental Rights: Equality (14–18) | Freedom (19–22) | Against Exploitation (23–24) | Freedom of Religion (25–28) | Cultural & Educational (29–30) | Constitutional Remedies (32)

🔑 Champakam Dorairajan Case 1951 → TN gave reservation → SC struck it down → Parliament added Clause 4 via 1st Amendment



⑩ Polity — Parliament's Power Over States 🌐

Q10. Under which Articles does Parliament have the power to create/alter states?

☐ A Articles 1 and 2 only

☐ B Article 368 only

☒ C Articles 2, 3, and 4 ✓

☐ D Articles 1, 2, 3, and 368

✓ Answer: C — Art.2 (new states), Art.3 (alter states), Art.4 (done by simple majority — NOT Art.368)

Article	Content
Article 1	"India, that is Bharat, shall be a Union of States"
Article 2	Admission/establishment of new states (admission of foreign territory into India)
Article 3	Formation/alteration of states — creating new; altering boundaries, names, areas
Article 4	Changes under Art.2 & 3 done by simple majority (NOT under Art.368)

INDIA vs USA

India: Indestructible Union of
Destructible States

BERUBARI CASE (1960)

To **acquire** territory → simple majority
(Art.3)

USA: Indestructible Union of

Indestructible States

States in India **CANNOT secede** from Union

To **cede** territory → **Art.368**

amendment required

Nehru-Noon Agreement (PM of Pak =

Feroz Khan Noon)

100th Amendment → enclaves to

Bangladesh

 MEMORY TRICK

Preamble: Berubari Case (1960) → Preamble NOT part of Constitution | Kesavananda

Bharati Case (1973) → Preamble IS part → CAN be amended



⑪ Sports — Long Jump Physics

Q11. Stmt 1: Athlete attempts maximum momentum by running before takeoff. | Stmt 2: During takeoff, athlete pushes off the weak foot to maintain balance.

☒ **A Only Statement 1 is correct ✓**

☐ ~~B Only Statement 2 is correct~~

☐ ~~C Both are correct~~

☐ ~~D Both are incorrect~~

✓ **Answer: A** — Statement 1 ✓ (running builds momentum) | Statement 2 ✗ (athlete pushes off **STRONGER** foot, not weak foot)

 KEY FACTS

 Statement 1 ✓ → Running builds momentum → longer jump (standing jump = shorter jump)

 Statement 2 ✗ → Athlete pushes off **STRONGER foot** for maximum force — NOT weak foot



⑫ Biology — Modes of Nutrition

Q12. Which organism breaks down food material **OUTSIDE** their bodies and then absorbs it?

- ☐ A ~~Animals like cows and lions~~
- ☐ B ~~Plants like grass and trees~~
- ☒ C **Fungi like bread mold, yeast, and mushrooms ✓**
- ☐ D ~~Parasites like Cuscuta and tapeworm~~

✓ Answer: C — Fungi (Saprophytic nutrition) = break down food **OUTSIDE** body using enzymes → then absorb

→ Modes of Nutrition

Type	How they feed	Examples
Autotrophic	Make own food (photosynthesis)	Green plants, blue-green algae, phytoplankton
Saprophytic	Break down food OUTSIDE body → absorb	Fungi (bread mold/Rhizopus, yeast, mushroom)
Parasitic	Extract from living host	Cuscuta (Amarbel), Plasmodium, Tapeworm
Holozoic	Ingest → digest INSIDE body → excrete	Humans, dogs, Amoeba

→ Fungi – Three Types

 Bread Mold (Rhizopus) Sporangia → Spore formation (asexual)	 Yeast Unicellular → Budding (asexual reproduction)
 Mushroom Multicellular fungi	

KEY FACTS

- 🔑 Spore formation = asexual reproduction → single parent → high genetic similarity, low variation
- 🔑 Amoeba (Holozoic): Pseudopods → food vacuole → digest inside → excrete
- 🔑 Phylum Platyhelminthes: Planaria (free-living, regeneration) | Liver Fluke (parasitic) | Tapeworm (parasitic)
- 🔑 Insectivorous plants: **Venus Flytrap, Bladderwort** (neither saprophytic nor parasitic)
- 🔑 Autotrophs = base/bottom of food pyramid = **Producers**

⑬ ⑭ Physics — Structure of the Sun ☀

Q13/14. Which layer of the Sun is primarily responsible for the formation of sunspots?

Ⓐ **Photosphere** ✓

Ⓑ Chromosphere

Ⓒ Corona

Ⓓ Core

✓ Answer: A — Photosphere = visible surface = sunspots form here | Asked in CHSL Mains recently!

→ Structure of the Sun

INNER LAYERS (core outward)

Core → Nuclear fusion; highest energy

Radiative Zone → Energy by radiation

Convective Zone → Energy by convection

OUTER LAYERS

Photosphere → Visible surface;
Sunspots here!

Chromosphere → Reddish layer above
photosphere

Corona → Outermost; normally invisible;
visible during solar eclipse; **CME** originate
here

KEY FACTS

☞ **Sunspots** = Dark regions on photosphere | Caused by intense magnetic activity |

Temperature is **LOWER** than surrounding photosphere (hence dark)

☞ **CME (Coronal Mass Ejections)** originate from **Corona** — NOT sunspots directly

☞ Sun composition: ~ **70% Hydrogen** + ~ **30% Helium**

→ Aditya-L1 Mission — Lagrange Points

Points	Type	Distance
L1, L2, L3	Collinear (straight line)	L1-L2: 1.5 million km L2-L3: 300 million km
L4, L5	Equilateral triangle with Sun + Earth	—
Aditya-	Placed at L1 point	Monitors Sun continuously



15 Economy — Ease of Doing Business Index

Q15. A: Global EODB Rankings gave maximum weightage to "Starting a Business." | R: "Starting a Business" was one of the 10 indicators.

- ☐ A Both A and R correct; R explains A
- ☐ B Both correct; R does NOT explain A
- ☐ C A correct; R incorrect
- ☒ D A incorrect; R correct ✓

✓ Answer: D — Assertion FALSE (Starting a Business ≠ highest weightage) | Reason TRUE (it IS one of 10 indicators)

→ 10 Indicators of Ease of Doing Business

1. Starting a Business
2. Getting Electricity
3. Getting Credit
4. Resolving Insolvency (India's IBC → 2016)
5. Enforcing Contracts
6. Paying Taxes
7. Protecting Minority Investors
8. Dealing with Construction Permits
9. Registering Property
10. Trading Across Borders

🔑 KEY FACTS

- 🔑 Released by: **World Bank** (currently discontinued/suspended)
- 🔑 World Bank + IMF = **Twin Organizations** | Created at **Bretton Woods Conference, 1944**
- 🔑 Both HQ: **Washington D.C., USA**



16 Biology — Cell Division: S-Phase

Q16. A cell has 46 chromosomes. How many chromatids will there be after the S-phase?

- ☐ A 46
- ☒ B 92 ✓
- ☐ C 23
- ☐ D 184

✓ Answer: B — After S-phase: each chromosome duplicates → $46 \times 2 = 92$ chromatids

→ Cell Cycle Phases

Phase	What Happens
G1	Cell grows
S (Synthesis)	DNA replication — DNA/chromatids DOUBLE
G2	Cell prepares for mitosis
M	Mitosis occurs (PMAT: Prophase, Metaphase, Anaphase, Telophase)

MITOSIS vs MEIOSIS

Mitosis: $1 \rightarrow 2$ daughter cells | Diploid ($2n$) | Somatic cells

Meiosis: $1 \rightarrow 4$ daughter cells | Haploid (n) | Sex cells (gametes)

CHROMOSOME FACTS

Composition: **DNA + Histone protein**

Location: **Nucleus**

Human: **46 chromosomes = 23 pairs**

22 pairs = Autosomal | 1 pair = Sex

(XX=♀, XY=♂)

◆ ◆ ◆

⑪ Medieval History — Chronological Order of Monuments

Q17. Arrange chronologically: Red Fort (Delhi) | Jahangir Mahal (Orchha) | Agra Fort

- ☐ A Red Fort → Agra Fort → Jahangir Mahal
- ☐ B Jahangir Mahal → Agra Fort → Red Fort
- ☒ C Agra Fort → Jahangir Mahal → Red Fort ✓
- ☐ D Agra Fort → Red Fort → Jahangir Mahal

✓ Answer: C — Agra Fort (1573) → Jahangir Mahal (1605) → Red Fort (1648)

Monument	Builder	Year	Location	Note
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Agra Fort	Akbar	~1573	Agra	UNESCO World Heritage Site (1983)
Jahangir Mahal, Orchha	Veer Singh (Bundela ruler — NOT Jahangir!)	~1605	Orchha, Bundelkhand, MP	Built to welcome Jahangir on first visit
Red Fort (Delhi)	Shah Jahan	1648	Delhi	Built in red sandstone

EXAM TRAP!

Jahangir Mahal was built by Bundela Hindu ruler Veer Singh — NOT by Jahangir himself! Commonly confused in exams!

MUGHAL DYNASTY ORDER

Babur → Humayun → Akbar → Jahangir → Shah Jahan → Aurangzeb



⑱ Sports — First Asian Olympic Gold in Javelin 🏆

Q18. Which Indian athlete became the first Asian to win gold in the javelin throw?

- ☒ A Neeraj Chopra ✓
- ☐ B Devendra Thajharia
- ☐ C Vikas Gowda
- ☐ D Seema Antil

✓ Answer: A — Neeraj Chopra | First Indian AND first Asian | 2020 Tokyo Olympics (held 2021) | Men's Javelin Throw



⑲ Geography — Northern Plains: Population Density 🏠

Q19. Which feature of the Northern Plains contributes to their very high population densities?

- ☐ A Presence of dense forests preventing other activities
- ☐ B Extreme cold weather suitable for specific crops
- ☒ C Flat terrain with fertile alluvial soil making farming easy ✓
- ☐ D Presence of large mineral deposits

✓ Answer: C — Flat terrain + Fertile alluvial soil = easy farming = high population density

→ Population Density — India

382

India Average
(persons/sq km)

1,106

Bihar
Highest State

11,320

Delhi
Highest UT

LOWEST DENSITY

Lowest State: **Arunachal Pradesh — 17**

Lowest UT: **Andaman & Nicobar Islands**

2nd Highest State: **West Bengal**

SUB-REGIONS OF NORTHERN PLAINS (N→S)

Bhabar (8–16 km): Rocky; rivers disappear underground

Terai (15–30 km): Marshy; rivers reappear; dense forests

Alluvial Plains: Bhangar (old) + **Khadar** (new, more fertile)

♦ ♦ ♦

②0 Medieval Literature — Prithviraj Raso

Q20. Who wrote the epic poem Prithviraj Raso?

- ☒ A Chandrabardai ✓
- ☐ B Kathan
- ☐ C Banabhatta
- ☐ D Amir Khusro

✓ Answer: A — Chandrabardai | Prithviraj Raso = love story of Prithviraj Chauhan III & Sanyogita

→ Important Medieval Works & Authors

Author	Work	About
Chandrabardai	Prithviraj Raso	Love story of Prithviraj Chauhan III + Sanyogita
Kalhan	Rajatarangini	History of Kashmir
Banabhatta	Harshacharita	Biography of Harshavardhan
Banabhatta	Kadambari	Novel
Amir Khusro	Nuh Sifar, Khaza-in-ul-Futuh, Tughlaqnama	Historical works; Title: "Tuti-e-Hind"
Malik Muhammad Jayasi	Padmavat	Story of Padmavati (wife of Raja Ratan Singh)

PRITHVIRAJ CHAUHAN

Abducted/married **Sanyogita** from swayamvara

Jaichand (Kannauj) angered → invited

Muhammad Ghori

1st Battle of Tarain (1191): Prithviraj defeated Ghori

2nd Battle of Tarain (1192): Ghori defeated Prithviraj

Battle of Chandawar (1194): Jaichand defeated

AMIR KHUSRO

Title: **Tuti-e-Hind** (Parrot of India)

Given by: **Alauddin Khilji**

Father of **Qawwali**

Invented: **Sitar + Tabla**

Harshavardhan's books: Ratnavali, Priyadarshika, Nagananda



Polity — Article for Judicial Review

Q21. The Supreme Court's power of judicial review primarily rises from which article?

- ☐ A Article 12
- ☒ B Article 13 ✓
- ☐ C Article 15
- ☐ D Article 17

✓ Answer: B — Article 13: laws inconsistent with FRs = void & null = primary source of Judicial Review

KEY FACTS

- 🔑 **Article 13**: Any law (Parliament/State Legislature) inconsistent with FRs = void | Includes bylaws, ordinances, regulations, pre-constitutional laws
- 🔑 **Constitutional amendments do NOT fall under Article 13** — basic structure doctrine (KBC case) applies separately
- 🔑 **Article 137**: Supreme Court has power to review its own judgments
- 🔑 **Kesavananda Bharati Case (1973)**: Established Basic Structure Doctrine | If amendment destroys basic structure → struck down



Geography — Barren Island

Q22. What is the significant feature of Barren Island in India?

- ☐ A Largest island in Andaman
- ☐ B Southernmost point of India
- ☒ C India's only active volcano ✓
- ☐ D India's only dormant volcano

✓ Answer: C — Barren Island = India's only ACTIVE volcano | Narcondam = only DORMANT volcano

BARREN ISLAND

Location: Between **Middle & South Andaman**

Feature: India's only **ACTIVE volcano**

NARCONDAM ISLAND

Location: **East of North Andaman**

Feature: India's only **DORMANT volcano**

Channel	Separates
Duncan Passage	South Andaman and Little Andaman
10° Channel	Little Andaman and Car Nicobar (Andaman from Nicobar)
9° Channel	Lakshadweep main islands and Minicoy
8° Channel	Minicoy and Maldives



Culture — Jawa Festival, Jharkhand 🎉

Q23. Which best reflects the symbolic core of the Jawa Festival celebrated by tribal girls in Jharkhand?

- ☐ A ~~Worship of ancestral spirits through fire rituals only~~
- ☐ B ~~Celebrates harvest season with community feasting~~
- ☒ C **Fertility ritual where sprouted seeds and melon symbolize fertility and offspring ✓**
- ☐ D ~~Commemorates victory in ancient tribal wars~~

✓ Answer: C — Jawa = fertility ritual | Sprouted seeds + melon = fertility & offspring | Celebrated by unmarried tribal girls

🧠 MEMORY TRICK

Memory Trick (Jharkhand Festivals): "Rohini's husband ran away, felt like peeing, did it at the border"

Mnemonic	Festival	Key Detail
Rohini	Rohini Festival	—
Ran away (Jawa=sprouting)	Jawa Festival	Sprouted seeds + melon; fertility; unmarried tribal girls
Pee (Karma)	Karma/Karam Festival	Karam tree/deity worshipped
Border (Bhagta Parb)	Bhagta Parb	Honors Buddha Baba (B=Bhagta, B=Buddha Baba)



Economy — MUDRA Yojana 💰

Q24. MUDRA Yojana provides credit support mainly for?

- ☐ A Large corporations and multinational companies
- ☐ B Agricultural sector only
- ☒ C Micro and Small Enterprises (MSMEs) ✓
- ☐ D Housing loans for government employees

✓ Answer: C — MUDRA = Micro Units Development and Refinance Agency | Collateral-free loans to MSMEs | Funded by SIDBI

→ MUDRA — Key Details

MUDRA FACTS

Full form: **Micro Units Development and Refinance Agency**

Launched: **April 8, 2015**

Loan type: **Collateral-free**

Funded by: **SIDBI (100%)**

SIDBI HQ: **Lucknow** | Est. 1990

FOUR LOAN CATEGORIES

Shishu → up to ₹50,000

Kishor → ₹50,000 to ₹5 lakh

Tarun → ₹5 lakh to ₹10 lakh

Tarun Plus (NEW!) → ₹10 lakh to ₹20 lakh

◆ ◆ ◆

Current Affairs — One Station One Product Scheme 🚂

Q25. What is the primary operational objective of the One Station One Product Scheme?

- ☐ A Converting ALL small and medium stations into large retail shopping malls
- ☐ B Mandating FSSAI certification for ALL food items SOLELY at railway stalls
- ☐ C Establishing duty-free trade zones for imported goods at all major stations
- ☒ D Providing livelihood support and welfare for local artistes and marginalized communities ✓

✓ Answer: D — Promotes local GI-tagged artisanal products; livelihood for local artisans & marginalized communities

 EXAM TIP

Option Elimination Strategy — Extreme Keywords:

- Options with "ALL", "NONE", "NEVER", "ALWAYS", "SOLELY" → usually WRONG
- Options describing complete conversion / abolition of something → usually WRONG
- Affirmative, positive, welfare-based statements → usually CORRECT
- Options A, B, C above have extreme/absurd keywords — eliminate immediately!



Quick Revision — All 25 Questions

Q	Topic	Answer
1	Sound wave Air → Water	C — Speed ↑, Wavelength ↑
2	Thermosphere characteristic	C — Temp can rise above 2000°C
3	Strongest acid in water	A — HCl (complete ionization)
4	Kuka Movement founder	B — Guru Ram Singh Ji (1857)
5	CAG borrowed from	A — Britain (Westminster model)
6	Temple architecture statements	B — Stmt 1 ✓, Stmt 2 ✗ (Vimana ≠ Gopuram)
7	Taekwondo matching	B (CBAD) — Tae=Feet, Kwon=Hand, Do=Way, Dojang=Hall
8	Nutrient-deficient tropical soil	D — Laterite Soil (leaching)
9	Article 33	B — Parliament can restrict FR for Armed Forces
10	Parliament power over states	C — Articles 2, 3, and 4
11	Long jump statements	A — Only Stmt 1 correct (stronger foot, not weak)
12	Breaks down food outside body	C — Fungi (Saprophytic nutrition)
13/14	Sunspot formation layer	A — Photosphere (CHSL Mains PYQ!)
15	Ease of Doing Business	D — Assertion wrong, Reason correct
16	Chromatids after S-phase	B — 92 (46 × 2, each chromosome duplicates)
17	Chronological order of monuments	C — Agra Fort → Jahangir Mahal → Red Fort
18	First Asian javelin Olympic gold	A — Neeraj Chopra (Tokyo 2020)
19	Northern Plains high density	C — Flat terrain + fertile alluvial soil
20	Author of Prithviraj Raso	A — Chandrabardai
21	Judicial review article	B — Article 13

22	Barren Island feature	C — India's only active volcano
23	Jawa Festival symbol	C — Fertility ritual, sprouted seeds
24	MUDRA Yojana for	C — Micro & Small Enterprises (MSMEs)
25	One Station One Product	D — Livelihood for local artisans



Topper Tips — Exam Ke Liye Yaad Rakho!



EXAM TIP

1. Sound in denser medium = **FASTER** /
Light in denser medium = **SLOWER**
2. Mesosphere = **COLDEST** /
Thermosphere = **HOTTEST**
(ionosphere)
3. Amalaka sits on Shikhara (Nagara) /
Vimana ≠ Gopuram (Dravidian)
4. After S-phase: 46 chromosomes → **92 chromatids**



EXAM TIP

5. CAG = from **Britain** / FR + Judicial Review = from **USA** / Concurrent List = **Australia**
6. Extreme keywords
(**all/none/never/solely**) → usually **WRONG** in options!
7. Matching questions: find **common pairs** between options first → eliminates 2 instantly
8. Khadar (new alluvium) > Bhangar (old alluvium) in fertility



END OF BLITZ MOCK #9 NOTES

All 25 Questions Covered ✓ | Mock karo, Revise karo, Top karo! 🎯

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